

William Lu

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Education

University of Illinois Urbana-Champaign
B.S. in Mathematics & Computer Science

Sep 2024 - Expected May 2027
GPA: 4.0/4.0

Previous study: University of Wisconsin-Madison, Computer Science (2023-2024) GPA: 4.0/4.0

Skills & Coursework

Languages: Mandarin (Native), English (Fluent), Japanese (Functional)

Technical Skills: C/C++, Python, Java, C#, JavaScript; PyTorch, CUDA, Django, Unity, LangGraph; Perfetto, Nsight Systems, Nsight Compute, ADB, RISC-V, Git

Relevant Coursework: Machine Learning, Applied Parallel Programming, Algorithms, Computer Systems

Publication: Comparisons of LSTM and other Machine Learning Approaches on Predicting Suicidal Tendency Regarding to Social Media Posts (2021)

Experience

TikTok - Effect Core SDK
Software Engineer Intern

San Jose, CA
May 2026 – Aug 2026

- Built an auto-trace collection-to-report platform (Python, Perfetto, ADB/UIAutomator), replacing a 10-step manual workflow with a single command and producing a 1,200-trace, 482-effect corpus across 6 device profiles; let two other engineers onboard new hardware and extend the collector without touching core logic.
- Designed the multi-layered agentic performance attribution and prioritization analysis tool behind it, giving engineers a ranked sense of which stage would yield the highest overall improvement if optimized. This project provided the data foundation for the Effects Performance Team's Q3 optimization roadmap.
- Established end-to-end performance instrumentation from high-level SDK down to individual JS functions in the effect package so engineers now get a complete picture inside a single Perfetto trace.

Projects

Convolutional Layer CUDA Kernel Optimization
CUDA Developer

Champaign, IL
Jan 2026 – May 2026

- Implemented and optimized convolution as CUDA matrix multiplication using Tensor Cores, shared-memory tiling, constant-memory caching, loop unrolling, CUDA streams, and mixed-precision computation, reducing combined convolution kernel time from 74.9 ms to 53 ms (29% reduction, 1.41× speedup) at batch size 10,000.
- Profiled GPU execution with Nsight Compute and Nsight Systems to analyze memory traffic, instruction mix, Tensor Core utilization, and bottlenecks; Tensor Core optimization alone reduced second-layer kernel time by 21%.

Job Application Management Web App
Backend Engineer

Champaign, IL
Sep 2025 – Dec 2025

- Built Django REST APIs and relational data models for a team-developed job application tracking platform, supporting CRUD operations across users, applications, companies, and related resources.
- Designed normalized SQLite schemas with foreign-key constraints and serializer validation to maintain relational integrity.

Self-Supervised Representation Learning with SimCLR (PyTorch)
Independent Developer

Champaign, IL
Aug 2025 – Oct 2025

- Reproduced the SimCLR self-supervised learning pipeline in PyTorch based on the original paper, implementing data augmentation, a ResNet encoder, projection head, NT-Xent contrastive loss, and downstream evaluation.
- Investigated training scalability and reproduction gaps under limited compute, experimenting with batch-size constraints, gradient accumulation, and efficient data loading while training on Google Colab.

Tiny-Chat
Programmer & Assistant

Champaign, IL
Aug 2025 – Oct 2025

- Contributed to a multi-agent conversational framework that orchestrates LLM-powered agents to simulate social interactions including negotiation, persuasion, and collaboration.
- Implemented environment initialization and agent-profile loading from local and Hugging Face datasets, along with structured conversation logging and JSON-based persistence for experiment records.

Unix-like Kernel
 Programmer & designer
 Champaign, IL
 May 2025 – Aug 2025

- Led a 3-person team in building a Unix-like kernel in C for RISC-V, implementing multitasking, virtual memory, a file system, system calls, and device drivers.
- Implemented a 2-way set-associative cache, process management, Unix pipes for IPC, and block/entropy device drivers.

Personal Website Development
 Independent Developer
 Personal
 Aug 2024 – Present

- Designed and built a personal website from scratch using HTML, CSS, and JavaScript.
- Deployed the site via GitHub Pages to demonstrate academic background, projects, and resume in a responsive, user-friendly portfolio format.

Minerva Game Studio
 Game Programmer
 Madison, WI
 Sep 2023 – Mar 2025

- Coded for a student-run studio to develop a stand-alone roguelike game using Unity Engine, whose demo is released on steam.
- Designed and built skills (ability of game characters) and assisted implementing language localization to enhance gameplay and improve player adaptability.

Cathaypath Institute of Science Research
 Research Group Leader
 Virtual
 Apr 2021 – Aug 2021

- Compared various Deep Learning models' ability in detecting one's mental health by analyzing social media posts.
- Led a student research group of 4 and co-published a paper on IEEE as a primary writer on Deep Learning's capacity in supporting people's mental health care.

Leadership

Association for Computing Machinery: SIGPwny
 Active Member
 Champaign, IL
 Sep 2024 – Mar 2025

- Practicing Cybersecurity and Hacking skills in a Special Interest Group that focuses on information security under the Association for Computing Machinery at UIUC.
- Cooperate with teammates in competing the Catch The Flag competitions using cybersecurity knowledge such as git, pwn, cryptography, etc.

Eduphoria EdTech Co. Ltd.
 General Assistant
 Madison, WI
 Sep 2023 – May 2024

- Organized events such as info sessions and ted talks in Madison campus for newly founded Chinese Company focusing on international education.
- Edited videos using CapCut for the Marketing Department, and collected data for the IT department in order to expand influence and provide better service for students.

National High School Game Academy
 Game Programmer
 Pittsburgh, PA
 Jun 2022 – Aug 2022

- Applied various disciplines of game design such as narrative, sound, art, and coding to make video games, which visualized the possibilities of Human Computer Interaction with Virtual Reality (VR) in the video game industry.
- Created 2 video games in teams: a remake of the classic arcade game Centipede and VR Archery game using Unity Engine and HTC VIVE pro 2 (headset).

Game Design Club
 Founder & Leader
 Wheaton, IL
 Sep 2021 – Jun 2023

- Established a student club that focused on creating stand-alone computer video games, and hosted weekly meeting in which taught Python, C#, and Unity to members.
- Created a 2D Maze Game, 2D Platformer Game (like Mario), and developed a 3D game demo that utilized OpenAI API in non-player characters (NPC) to explore AI's capabilities to make NPCs more realistic and make the game more immersive.